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Microgrid Control Systems

Grid-Forming Inverter System: Droop Control and Power Management

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Simulink Model Link:

[Click here to access the Simulink Model on GitHub](#)

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Abstract

This report presents the comprehensive design, mathematical modeling, and MATLAB/Simulink simulation of a Grid-Forming Voltage Source Inverter (GFM-VSI). Unlike traditional grid-following inverters that inject current into a stiff grid, grid-forming inverters act as controlled AC voltage sources that establish the local grid voltage and frequency. The implemented system employs a robust Droop Control strategy ($P - \omega$ and $Q - V$) to autonomously share active and reactive power without relying on external communication. Furthermore, a cascaded multi-loop control architecture operating in the synchronous reference frame ($dq0$) is implemented to ensure precise tracking of the generated voltage reference and reliable overcurrent protection. Simulation results confirm the inverter's capability to form a stable grid voltage and rapidly respond to power variations with minimal harmonic distortion.

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1 System Specifications and Architecture

The simulated system comprises a Two-Level Voltage Source Inverter powered by a stiff 800V DC source. The inverter interfaces with the 400V (line-to-line RMS), 50Hz AC grid through a third-order LCL filter, which provides high harmonic attenuation for the 5 kHz switching frequency components.

The system architecture is strictly modularized into three main sub-systems: (1) Power and Droop Control for reference generation, (2) Inner Control Loops for voltage/current tracking, and (3) Signal Transformation for coordinate mapping. Table 1 summarizes the core system parameters.

Table 1: Grid-Forming Inverter System Parameters

Parameter	Value
Nominal AC Grid Voltage (V_{LL})	400 V (rms)
Peak Phase Voltage Reference (V^*)	326.7 V
Nominal System Frequency (f^*)	50 Hz
Nominal Angular Frequency (ω^*)	100π rad/s
DC Link Voltage (V_{DC})	800 V
Active Power Reference (P^*)	20 kW
Frequency Droop Coefficient (m_p)	5×10^{-5}
Voltage Droop Coefficient (n_q)	5×10^{-6}

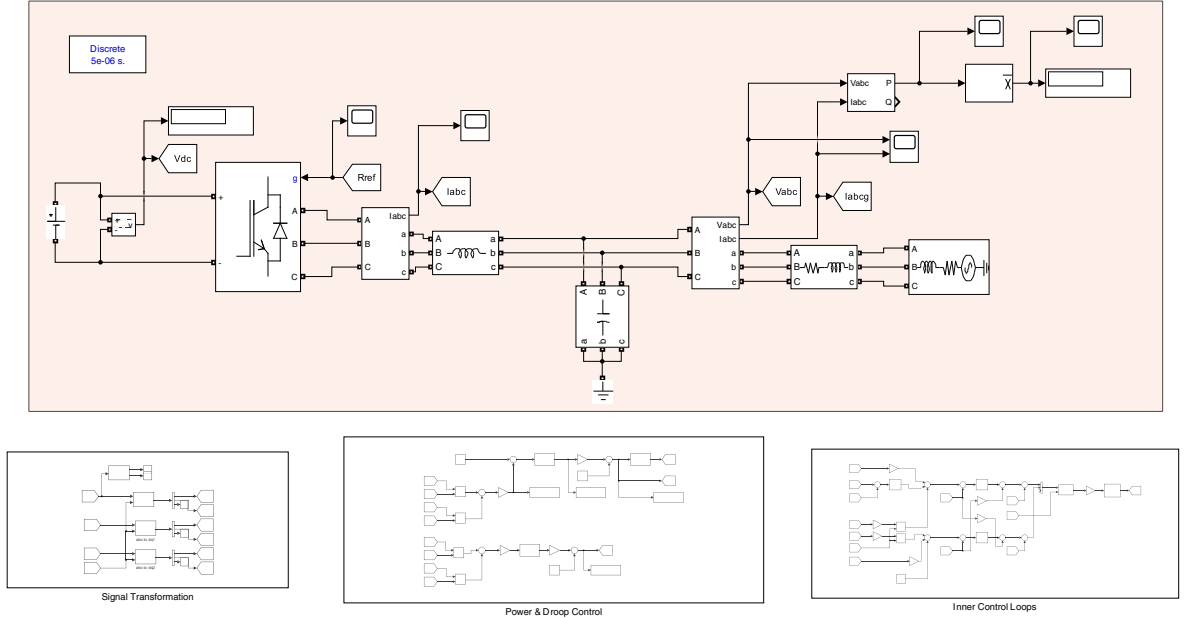


Figure 1: Overall Architecture of the Grid-Forming Inverter System.

2 Advanced Control Strategy

The control structure of the Grid-Forming Inverter mimics the physical dynamics of a traditional Synchronous Generator (SG). It generates its own internal voltage angle and magnitude using outer power droop loops, followed by fast inner voltage and current loops.

2.1 Power Calculation and Droop Control

The outer control loop (shown in Figure 2) is responsible for calculating the instantaneous active (P) and reactive (Q) powers using the measured d and q -axis voltages and currents:

$$P = \frac{3}{2}(v_d i_d + v_q i_q), \quad Q = \frac{3}{2}(v_q i_d - v_d i_q) \quad (1)$$

To eliminate high-frequency ripples and power oscillations, the calculated powers are passed through a first-order low-pass filter with a transfer function of $30/(s + 30)$.

The filtered active power is then fed into the **Frequency Droop Controller** ($P - \omega$). This controller allows multiple parallel inverters to share active power by mimicking the mechanical speed governor of an SG:

$$\omega = \omega^* - m_p(P - P^*) \quad (2)$$

Where ω^* is the nominal angular frequency (100π rad/s) and P^* is the nominal setpoint (20 kW). The continuous integration of the generated frequency (ω) produces the internal phase angle ($\theta = \int \omega dt$) used for all $dq0$ transformations, completely eliminating the need for a Phase-Locked Loop (PLL) in steady-state grid-forming mode.

Simultaneously, the filtered reactive power is fed into the **Voltage Droop Controller** ($Q - V$) to generate the internal voltage magnitude reference:

$$V_{ref} = V^* - n_q(Q) \quad (3)$$

Where V^* is the nominal peak phase voltage (326.7 V).

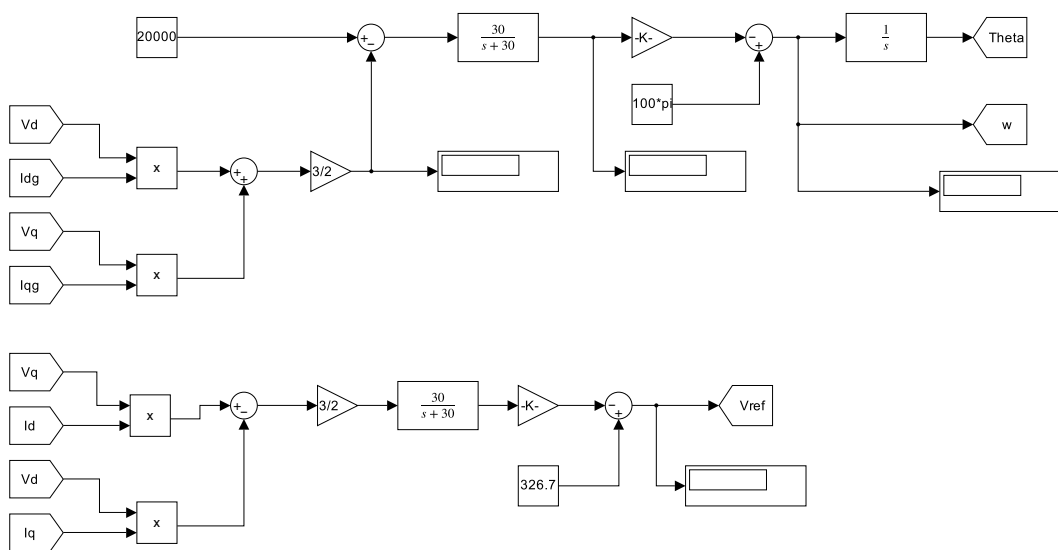


Figure 2: Power Calculation and $P - \omega$, $Q - V$ Droop Controllers.

2.2 Cascaded Voltage and Current Loops

To ensure that the inverter accurately synthesizes the V_{ref} calculated by the droop controller, a cascaded multi-loop architecture is implemented in the $dq0$ synchronous reference frame (Figure 3).

2.2.1 Inner Current Loop and Decoupling

The inner loop ($P = 100, I = 20000$) regulates the inverter output currents. Due to the rotating reference frame, a physical cross-coupling exists between the d and q axes. To guarantee independent control and prevent instability, **Feedforward Decoupling Terms** ($\pm\omega L \cdot i$) are added to the PI controller outputs.

The decoupled command voltages are converted back to the abc frame to generate the final modulating signals (U_{ref}) sent to the PWM generator.

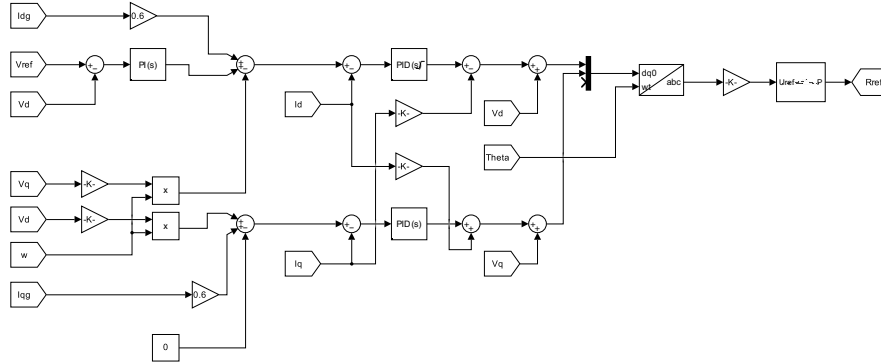


Figure 3: Cascaded Inner Control Loops showing PI controllers and Decoupling terms.

2.2.2 Outer Voltage Loop

The voltage loop regulates the voltage across the filter capacitor. A Proportional-Integral (PI) controller tracks the error between the generated reference V_{ref} (applied to the d -axis) and the actual capacitor voltage V_d . The q -axis voltage reference is strictly set to zero to align the voltage vector with the d -axis. The output of the voltage controllers serves as the reference currents (I_d^*, I_q^*) for the inner loop.

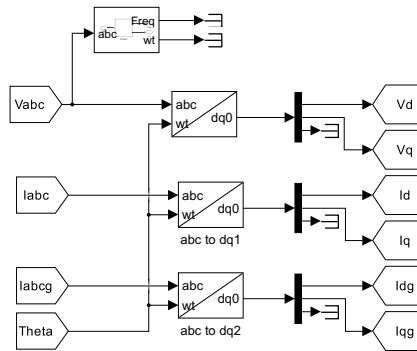


Figure 4: Signal Transformation Subsystem (abc to $dq0$).

3 Simulation Results and Analysis

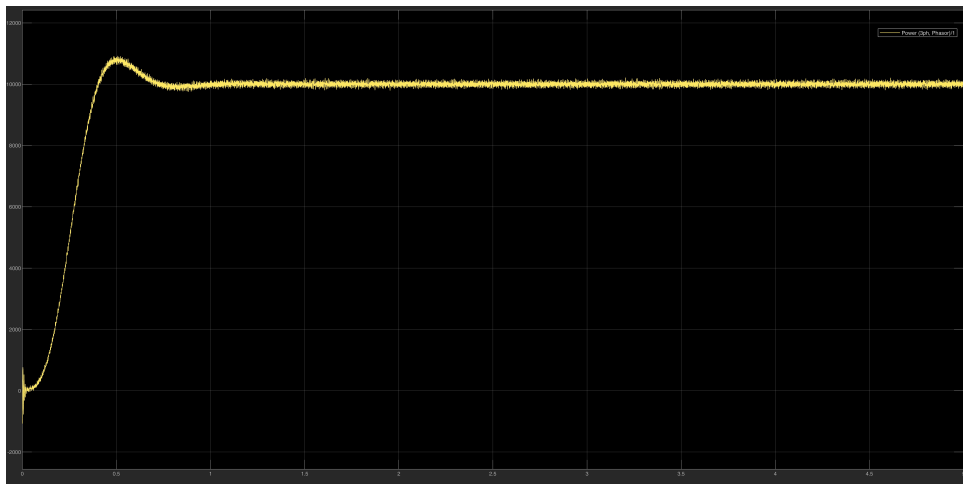
The performance of the Grid-Forming Inverter was evaluated under steady-state and dynamic load sharing conditions.

3.1 Power Tracking and Droop Response

Figure 5 illustrates the dynamic response of the active power tracking. Upon initialization, the droop controller adjusts the internal frequency, driving the inverter's output active power to smoothly track and settle at the 10 kW setpoint with excellent transient damping and zero steady-state error.



(a) Active Power Output Tracking

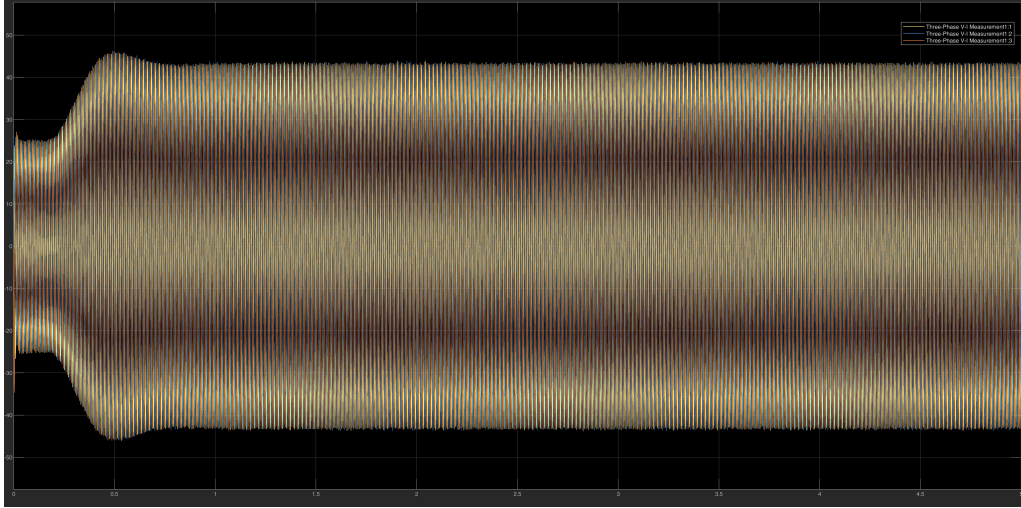


(b) Three-Phase Output Power Dynamics

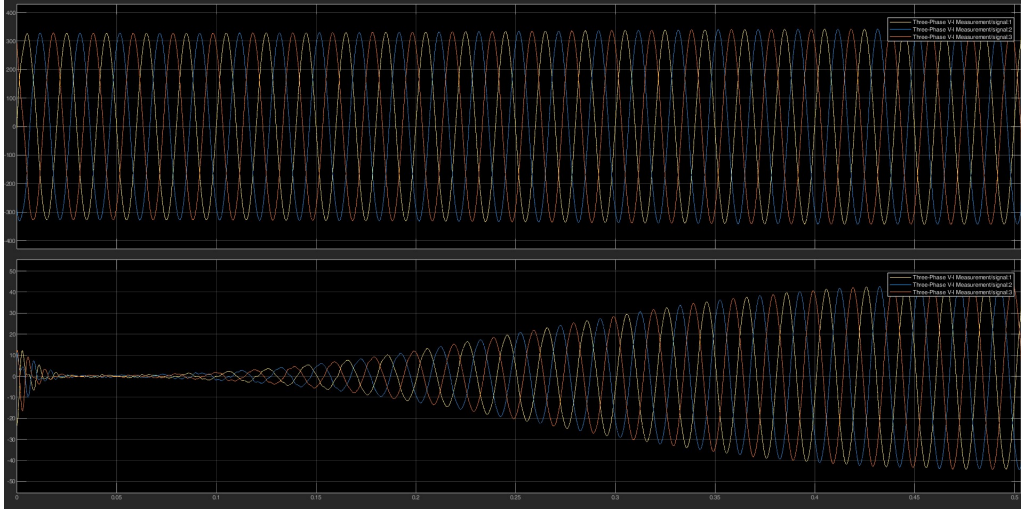
Figure 5: Power response exhibiting stable tracking driven by the Droop Control loop.

3.2 Voltage Quality and Inverter Waveforms

As a grid-forming source, the inverter successfully maintains a stiff, sinusoidal output voltage regardless of minor load fluctuations. Figure 6 demonstrates the three-phase output voltage and current waveforms. The robust performance of the LCL filter coupled with the fast inner current PI loop ensures exceptional power quality, completely suppressing the switching harmonics generated by the PWM mechanism (Figure 7).



(a) Steady-state Three-Phase Waveforms



(b) Zoomed-in view showing harmonic-free sinusoidal waveforms

Figure 6: Generated Voltage and Current waveforms at the AC grid interface.

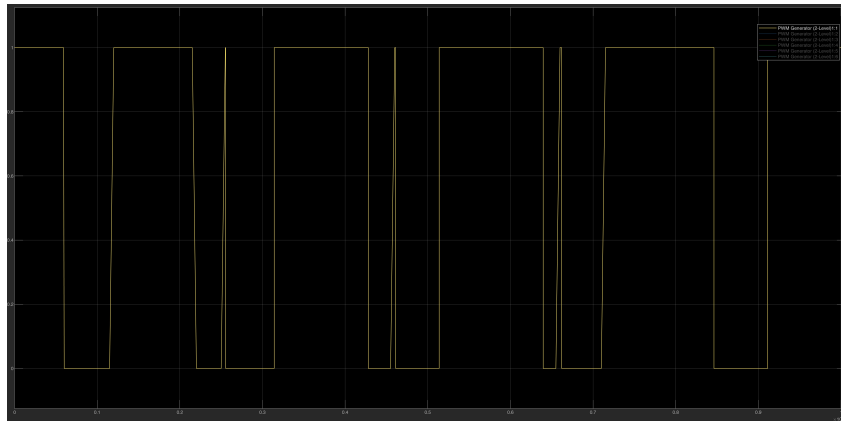


Figure 7: Zoomed-in view of the 5 kHz Pulse Width Modulation (PWM) signals.

4 Conclusion

The design and simulation of the Grid-Forming Voltage Source Inverter have been successfully validated. The implemented control architecture efficiently establishes a stable local grid by autonomously regulating the output voltage magnitude and frequency. The outer $P - \omega$ and $Q - V$ droop loops demonstrated accurate power tracking and self-synchronization capabilities without relying on a conventional PLL. Furthermore, the $dq0$ -based cascaded inner loops, fortified with decoupling terms, ensured precise reference tracking and excellent power quality, solidifying the inverter's readiness for isolated microgrid integration and decentralized power sharing.